Ari Rabinovitz

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Education

Cornell University, B.S. in Mechanical Engineering, *summa cum laude* *May 2026*

GPA:	4.08/4.30
Honors:	Dean's List 2022-2026
	Tau Beta Phi, Engineering Honor Society 2024-2026
Relevant coursework:	Spacecraft Technology and System Architecture, Intermediate Dynamics, Mechanics of Materials, Heat Transfer, Fluid Mechanics, Statics and Mechanics of Solids, Combustion Processes, Intro to Spaceflight Mechanics, Dynamics, Propulsion of Aircraft and Rockets

Skills

- CAD: SolidWorks, Creo, Autodesk Inventor, Autodesk Fusion 360
- Computer Languages: MATLAB, C, Python
- Finite Element Analysis: ANSYS, SolidWorks Simulation
- Other: 3-D Printing, GD&T, Mechanical Drawing, DFMA, Soldering, Mill, Lathe, Microsoft Excel

Relevant Experience

***Structural Mechanics and Concepts Intern*, NASA Langley Research Center** *June 2026 – Present*

- Design latching mechanism for Lunar System Manipulator System (LSMS) to carry payloads of up to 1000 kg on the lunar surface.
- Develop end effector for LSMS designed to carry loads on uneven terrain, utilizing capstan mechanisms.
- Prototype joints to enable autonomous robotic construction of lunar surface structures.

***Deployable Technologies Intern*, Physical Sciences, Inc.** *June 2025 – Aug. 2025*

- Designed and iterated on a quick-release gripper using SolidWorks and 3D-printed prototypes, each capable of supporting 165 lbs., and integrated into a rescue litter carry device intended for long-distance transports.
- Demonstrated satisfaction of gripper's design requirements through experimental tests and finite element analysis (SolidWorks simulation) with mass-production gripper material.
- Integrated sensors to test fuel pump intended for use with low lubricity fluids.
- Assisted in ideation of mobile lightweight folding bridge that can deploy and be retrieved in under 5 minutes and span a 15-meter gap while maintaining a load of 60 tons.

***Lead, Arm sub-team*, Cornell Mars Rover Project Team** *Apr. 2025 – May 2026*
***Member, Arm sub-team*, Cornell Mars Rover Project Team** *Oct. 2022 – June 2025*

- Designed end effector and structural joints for a 6-axis robotic manipulator assigned to perform complex tasks including typing, flipping switches, picking up rocks, and securing a door with a screw.
- Reconfigured locations of motor and gearbox joints on arm to reduce the moment on the base gearbox by 12 percent.
- Created unified self-propelling screwdriver mechanism that actuates and rotates simultaneously.

***Research Assistant*, Cornell ASTRA Lab** *June 2024 – Aug. 2024*

- Conducted research of high-thrust electrospray propulsion technology for very low earth orbit (VLEO) satellites to allow operation for months to years instead of for days to weeks.
- Tested the firing of ionic and novel nanoparticle fluids to determine optimal propellant.
- Designed and milled components to add goniometers to the testing rig to support more precise testing.
- Built current sensing circuit that could measure up to 10 kV.

***Research Assistant*, Cornell Space Structures Lab** *Jan. 2024 – May 2025*

- Conducted research to manufacture lightweight, easily transportable, and durable structures in space.
- Developed deployable boom and shaping mechanism to enable in-space manufacturing while preventing buckling or blossoming of the boom.

Personal Interests

- Tennis, board games, hiking